

Quantum-Inspired Cyber Threat Detection

for teleportation-based Quantum Digital Signature systems

Core problem

Classical public-key signatures such as RSA and ECC depend on hardness assumptions that become vulnerable to sufficiently powerful quantum computers. The proposed system replaces that trust model with quantum evidence: unknown qubits cannot be copied perfectly, and unauthorized measurement leaves disturbance that can be verified later.

Project stance: deterministic physics checks only. No AI/ML classifier is required; every decision comes from QBER, mismatch rates, verifier asymmetry, freshness, and authentication fields.

Threats covered

Eavesdropping / tampering

External signature forgery

Dishonest verifier forgery

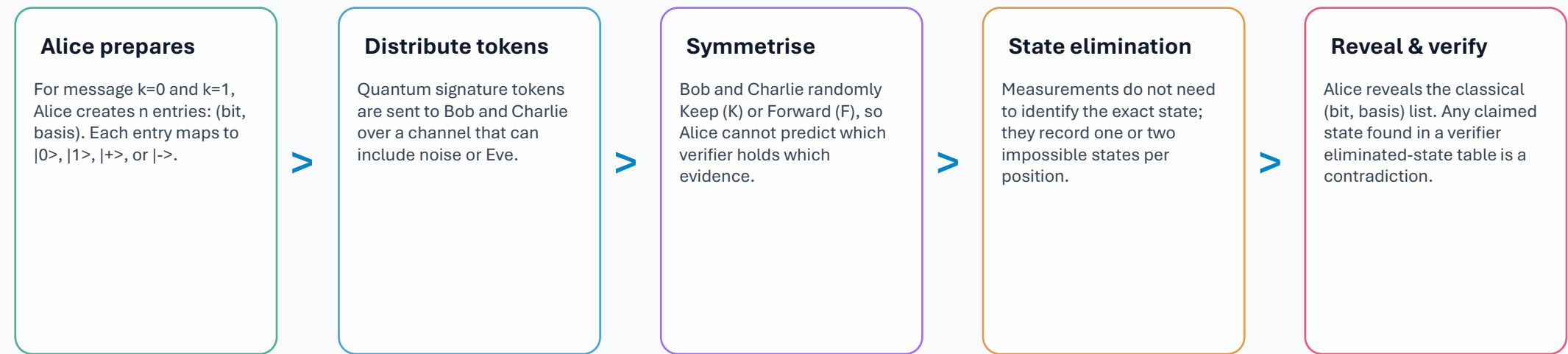
Repudiation between Bob and Charlie

Replay and sender impersonation

Output: a Qiskit/Aer simulation framework, adversary testbed, deterministic threat engine, security-math module, socket-network demo, and real-time telemetry dashboard.

QDS mechanism: signature becomes quantum evidence

Alice signs with BB84-style qubit tokens; verifiers store eliminated-state evidence before the classical signature is revealed.



BB84 token map

Bit	Basis	Gate	State
0	Z	identity	$ 0\rangle$
1	Z	X	$ 1\rangle$
0	X	H	$ +\rangle$
1	X	X then H	$ -\rangle$

Eliminated-state rule

Z basis outcome 0 eliminates $|1\rangle$; Z basis outcome 1 eliminates $|0\rangle$.
X basis outcome 0 eliminates $|-\rangle$; X basis outcome 1 eliminates $|+\rangle$.
Verification is based on contradictions against this stored evidence.

Basic QDS: mismatches == 0. Teleportation QDS: observed mismatch rate <= acceptance threshold s_a .

Eavesdropping detection: measurement disturbance becomes evidence

Intercept-resend is modeled as Eve measuring the qubit mid-channel and forwarding a possibly disturbed state.

Intercept-resend attack path

- 1 Eve intercepts each qubit with probability $p_{\text{intercept}}$.
- 2 Eve randomly chooses Z or X measurement basis.
- 3 The qubit is measured into Eve's classical register.
- 4 If Eve used the wrong basis, measurement collapses and disturbs the state.
- 5 Bob/Charlie later observe elevated QBER or QDS contradictions.

Why full intercept-resend gives about 25% QBER

Among sifted BB84 positions, Bob kept only cases where his basis matched Alice. Eve still has a 50% chance of choosing the wrong basis. When she is wrong, Bob has a 50% chance of receiving the wrong bit.

$$0.5 \text{ wrong basis} * 0.5 \text{ wrong bit} = 0.25 \text{ QBER}$$

Detection thresholds used in the project

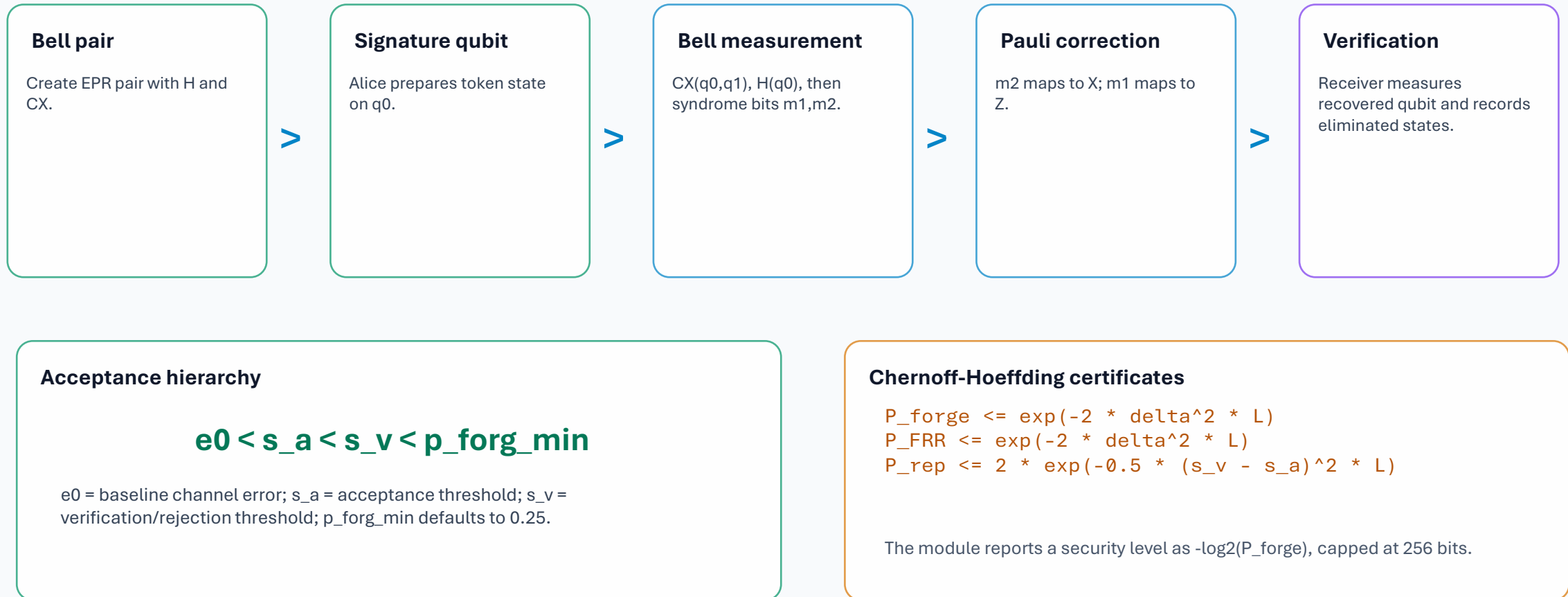
Communication layer: PointToPointProtocol and SecurePointToPointProtocol flag Eve when $qber > qber_threshold$; default threshold is 0.15.

QDS detection engine: channel eavesdropping is classified when $channel_qber > 0.11$, returning EAVESDROPPING_TAMPERING.

Signature-evidence view: Eve disturbance also increases Bob/Charlie mismatch rates and may create verifier asymmetry.

Teleportation-based QDS and statistical security model

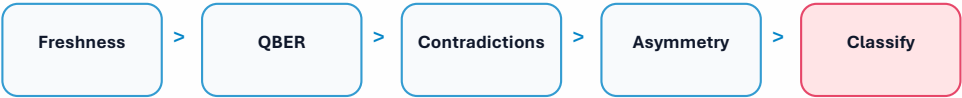
Stage 5 extends prepare-and-measure QDS with Bell pairs, syndrome recording, Pauli correction, and threshold certificates.



End-to-end system: deterministic threat engine and deliverables

The implementation combines a simulator, adversarial channel, deterministic classifier, security math, and network telemetry.

Decision pipeline



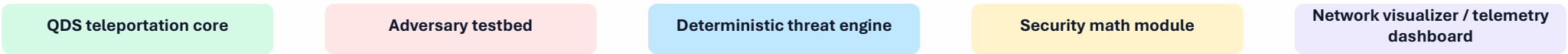
Priority order: missing tokens -> replay -> impersonation -> channel_qber > 0.11 -> clean authentic -> dishonest verifier -> repudiation -> channel noise -> external forgery.

Socket-network demo

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ChannelProxyNode : port 8000
AliceDaemon : port 8001
Bob / Charlie : ports 8002 and 8003
DetectorDaemon : port 8004
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In socket mode, qubits are represented as [bit, basis] descriptors. Eve modifies descriptors to mirror circuit-level measurement disturbance.

Expected deliverables



Key limitation: this is a simulator, not a deployed quantum network. The eavesdropping model is intercept-resend and the socket demo passes classical descriptors rather than physical qubits.